

# CS471 Lab 1

|              |                    |
|--------------|--------------------|
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## Part 1: Capturing HTTP Traffic

### Screenshot 1 – HTTP Packets

| *شبكة Wi-Fi                                                                |              |                          |                          |          |        |                                            |  |
|----------------------------------------------------------------------------|--------------|--------------------------|--------------------------|----------|--------|--------------------------------------------|--|
| File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help |              |                          |                          |          |        |                                            |  |
| http                                                                       |              |                          |                          |          |        |                                            |  |
| No.                                                                        | Time         | Source                   | Destination              | Protocol | Length | Info                                       |  |
| 1119                                                                       | 29.156984900 | 2001:16a2:3571:e200::... | 2600:1f13:37c:1400::...  | HTTP     | 501    | GET / HTTP/1.1                             |  |
| 1140                                                                       | 29.381384600 | 2600:1f13:37c:1400::...  | 2001:16a2:3571:e200::... | HTTP     | 1047   | HTTP/1.1 200 OK (text/html)                |  |
| 1143                                                                       | 29.389929200 | 2001:16a2:3571:e200::... | 2600:1f13:37c:1400::...  | HTTP     | 568    | GET /online HTTP/1.1                       |  |
| 1167                                                                       | 29.625322000 | 2600:1f13:37c:1400::...  | 2001:16a2:3571:e200::... | HTTP     | 667    | HTTP/1.1 301 Moved Permanently (text/html) |  |
| 1170                                                                       | 29.682045700 | 2001:16a2:3571:e200::... | 2600:1f13:37c:1400::...  | HTTP     | 569    | GET /online/ HTTP/1.1                      |  |
| 1171                                                                       | 29.934447600 | 2600:1f13:37c:1400::...  | 2001:16a2:3571:e200::... | HTTP     | 293    | HTTP/1.1 200 OK (text/html)                |  |
|                                                                            |              |                          |                          |          | 509    | GET /favicon.ico HTTP/1.1                  |  |
|                                                                            |              |                          |                          |          | 490    | HTTP/1.1 200 OK (PNG)                      |  |

|                                                                                         |  |      |                                                 |            |
|-----------------------------------------------------------------------------------------|--|------|-------------------------------------------------|------------|
| > Frame 1007: Packet, 501 bytes on wire (4008 bits), 501 bytes captured on interface... |  | 0000 | 2c ab 00 77 07 e1 68 3e 26 3f 71 6d 86 dd 60 07 | ,..w..h>   |
| > Ethernet II, Src: Intel_3f:71:6d (68:3e:26:3f:71:6d), Dst: Huawei...                  |  | 0010 | ac df 01 bf 06 40 20 01 16 a2 35 71 e2 00 ac 16 | ....@..    |
| > Internet Protocol Version 6, Src: 2001:16a2:3571:e200:ac16:2ef0:1c...                 |  | 0020 | 2e f0 1c ff d2 a7 26 00 1f 13 03 7c 14 00 ba 21 | .....&.    |
| > Transmission Control Protocol, Src Port: 55596, Dst Port: 80, Seq...                  |  | 0030 | 71 65 5f c7 73 6e d9 2c 00 50 ad d5 c8 5e 0f d6 | qe_..sn.,  |
| > Hypertext Transfer Protocol                                                           |  | 0040 | c8 4e 50 18 ff ff 75 d4 00 00 47 45 54 20 2f 20 | ..NP....u. |
|                                                                                         |  | 0050 | 48 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 | HTTP/1.1   |
|                                                                                         |  | 0060 | 6e 65 76 65 72 73 73 6c 2e 63 6f 6d 0d 0a 43 6f | neverssl   |
|                                                                                         |  | 0070 | 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2d 61 | nnection   |
|                                                                                         |  | 0080 | 6c 69 76 65 0d 0a 55 70 67 72 61 64 65 2d 49 6e | live..Up   |
|                                                                                         |  | 0090 | 73 65 63 75 72 65 2d 52 65 71 75 65 73 74 73 3a | secure-R   |
|                                                                                         |  | 00a0 | 20 31 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 | 1..User    |
|                                                                                         |  | 00b0 | 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 57 69 6e | Mozilla/   |
|                                                                                         |  | 00c0 | 64 6f 77 73 20 4e 54 20 31 30 2e 30 3b 20 57 69 | dows NT    |
|                                                                                         |  | 00d0 | 6e 36 34 3b 20 78 36 34 29 20 41 70 70 6c 65 57 | n64; x64   |
|                                                                                         |  | 00e0 | 65 62 4b 69 74 2f 35 33 37 2e 33 36 20 28 4b 48 | ebKit/53   |
|                                                                                         |  | 00f0 | 54 4d 4c 2c 20 6c 69 6b 65 20 47 65 63 6b 6f 29 | TML, lik   |
|                                                                                         |  | 0100 | 20 42 00 73 66 64 65 2f 31 3f 30 2e 20 2e 20 2e | ..h..h..   |

Hypertext Transfer Protocol: Protocol

Packets: 1433 · Displayed: 8 (0.6%) · Dropped: 0 (0.0%) Profile: Default

## Part 2: Analyzing TCP/IP Traffic

### Screenshot 2 – TCP Three-Way Handshake

The screenshot displays a Wireshark packet capture of a TCP three-way handshake. The packet list shows the following frames:

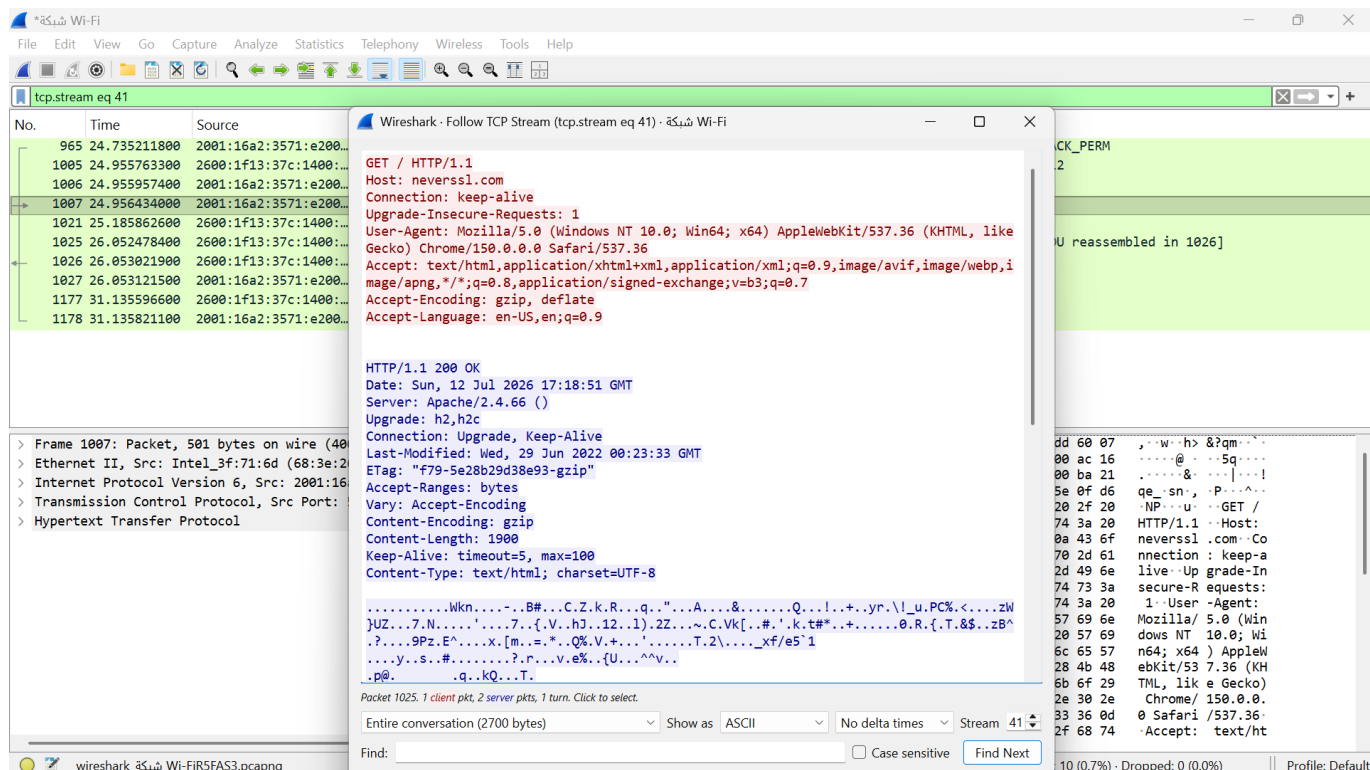
| No.  | Time         | Source                   | Destination              | Protocol | Length | Info                                                       |
|------|--------------|--------------------------|--------------------------|----------|--------|------------------------------------------------------------|
| 965  | 24.735211800 | 2001:16a2:3571:e200::... | 2600:1f13:37c:1400::...  | TCP      | 86     | 55596 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=256 SAC |
| 1005 | 24.955763300 | 2600:1f13:37c:1400::...  | 2001:16a2:3571:e200::... | TCP      | 78     | 80 → 55596 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1412 |
| 1006 | 24.955957400 | 2001:16a2:3571:e200::... | 2600:1f13:37c:1400::...  | TCP      | 74     | 55596 → 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0               |
| 1007 | 24.956434000 | 2001:16a2:3571:e200::... | 2600:1f13:37c:1400::...  | HTTP     | 501    | GET / HTTP/1.1                                             |
| 1021 | 25.185862600 | 2600:1f13:37c:1400::...  | 2001:16a2:3571:e200::... | TCP      | 74     | 80 → 55596 [ACK] Seq=1 Ack=428 Win=26800 Len=0             |
| 1025 | 26.052478400 | 2600:1f13:37c:1400::...  | 2001:16a2:3571:e200::... | TCP      | 1374   | 80 → 55596 [ACK] Seq=1 Ack=428 Win=26800 Len=1300 [TCP PDU |
| 1026 | 26.053021900 | 2600:1f13:37c:1400::...  | 2001:16a2:3571:e200::... | HTTP     | 1047   | HTTP/1.1 200 OK (text/html)                                |
| 1027 | 26.053121500 | 2001:16a2:3571:e200::... | 2600:1f13:37c:1400::...  | TCP      | 74     | 55596 → 80 [ACK] Seq=428 Ack=2274 Win=65535 Len=0          |
| 1177 | 31.135596600 | 2600:1f13:37c:1400::...  | 2001:16a2:3571:e200::... | TCP      | 74     | 80 → 55596 [FIN, ACK] Seq=2274 Ack=428 Win=26800 Len=0     |
| 1178 | 31.135821100 | 2001:16a2:3571:e200::... | 2600:1f13:37c:1400::...  | TCP      | 74     | 55596 → 80 [ACK] Seq=428 Ack=2275 Win=65535 Len=0          |

The packet details pane for frame 1005 shows the following structure:

- Frame 1005: Packet, 78 bytes on wire (624 bits), 78 bytes captured
- Ethernet II, Src: HuaweiTechno\_77:07:e1 (2c:ab:00:77:07:e1), Dst:
- Internet Protocol Version 6, Src: 2600:1f13:37c:1400:ba21:7165:5f...
- Transmission Control Protocol, Src Port: 80, Dst Port: 55596, Seq:

The packet bytes pane shows the raw data of the packet, including the Ethernet II header, the IPv6 header, and the TCP header.

## Screenshot 3 – TCP Data Transfer



This screenshot shows a Wireshark capture of a TCP stream (tcp.stream eq 41). The packet list on the left shows several packets, with packet 1007 selected. The packet details pane on the right shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 6, and Transmission Control Protocol. The packet bytes pane on the right shows the raw data of the selected packet, which is a GET request for / HTTP/1.1.

**Wireshark - Follow TCP Stream (tcp.stream eq 41) - شبكة Wi-Fi**

GET / HTTP/1.1  
Host: neverssl.com  
Connection: keep-alive  
Upgrade-Insecure-Requests: 1  
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36  
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,\*/\*;q=0.8,application/signed-exchange;v=b3;q=0.7  
Accept-Encoding: gzip, deflate  
Accept-Language: en-US,en;q=0.9

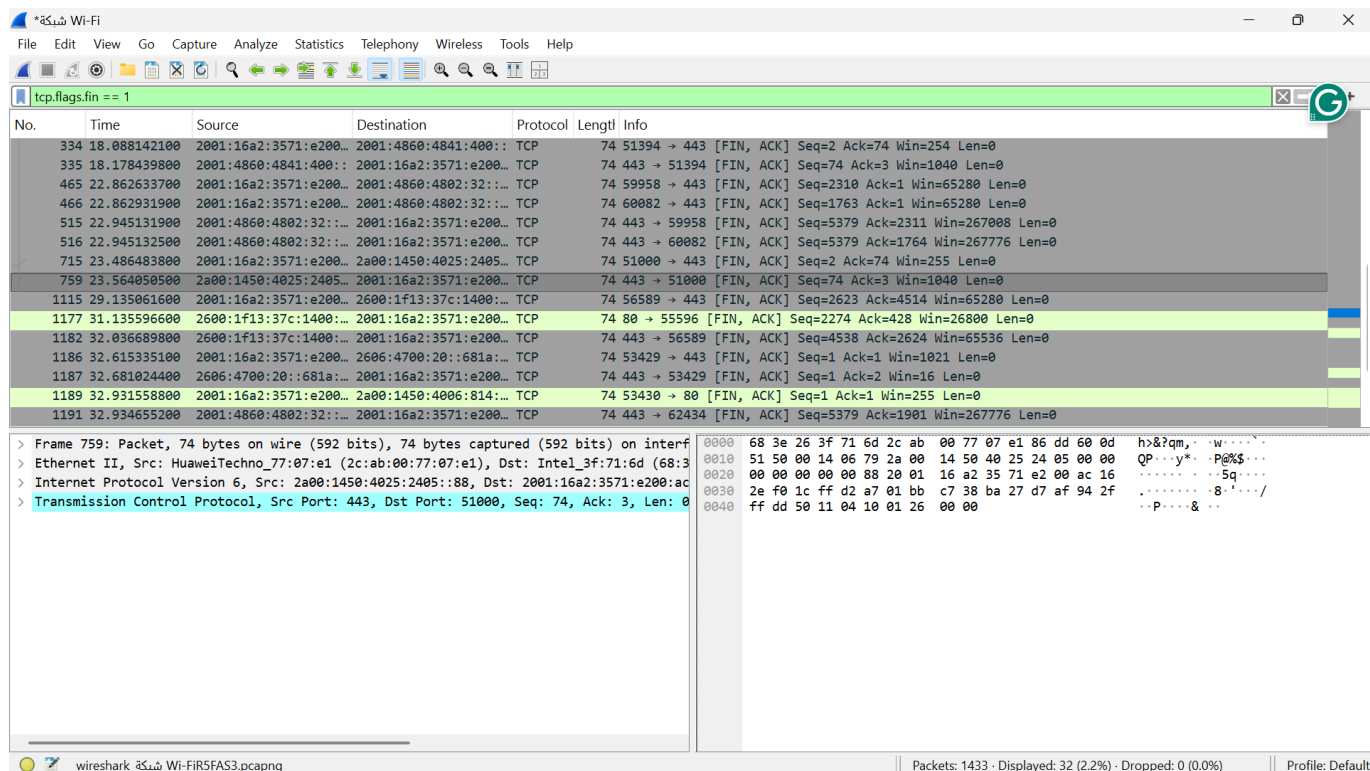
HTTP/1.1 200 OK  
Date: Sun, 12 Jul 2026 17:18:51 GMT  
Server: Apache/2.4.66 ()  
Upgrade: h2,h2c  
Connection: Upgrade, Keep-Alive  
Last-Modified: Wed, 29 Jun 2022 00:23:33 GMT  
ETag: "f79-5e28b29d38e93-gzip"  
Accept-Ranges: bytes  
Vary: Accept-Encoding  
Content-Encoding: gzip  
Content-Length: 1900  
Keep-Alive: timeout=5, max=100  
Content-Type: text/html; charset=UTF-8

Packet 1025: 1 client pkt, 2 server pkts, 1 turn. Click to select.

Entire conversation (2700 bytes) Show as ASCII No delta times Stream 41

Find: Case sensitive Find Next

## Screenshot 4 – TCP Termination



This screenshot shows a Wireshark capture of a TCP stream (tcp.flags.fin == 1). The packet list on the left shows several packets, with packet 759 selected. The packet details pane on the right shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 6, and Transmission Control Protocol. The packet bytes pane on the right shows the raw data of the selected packet, which is a FIN packet.

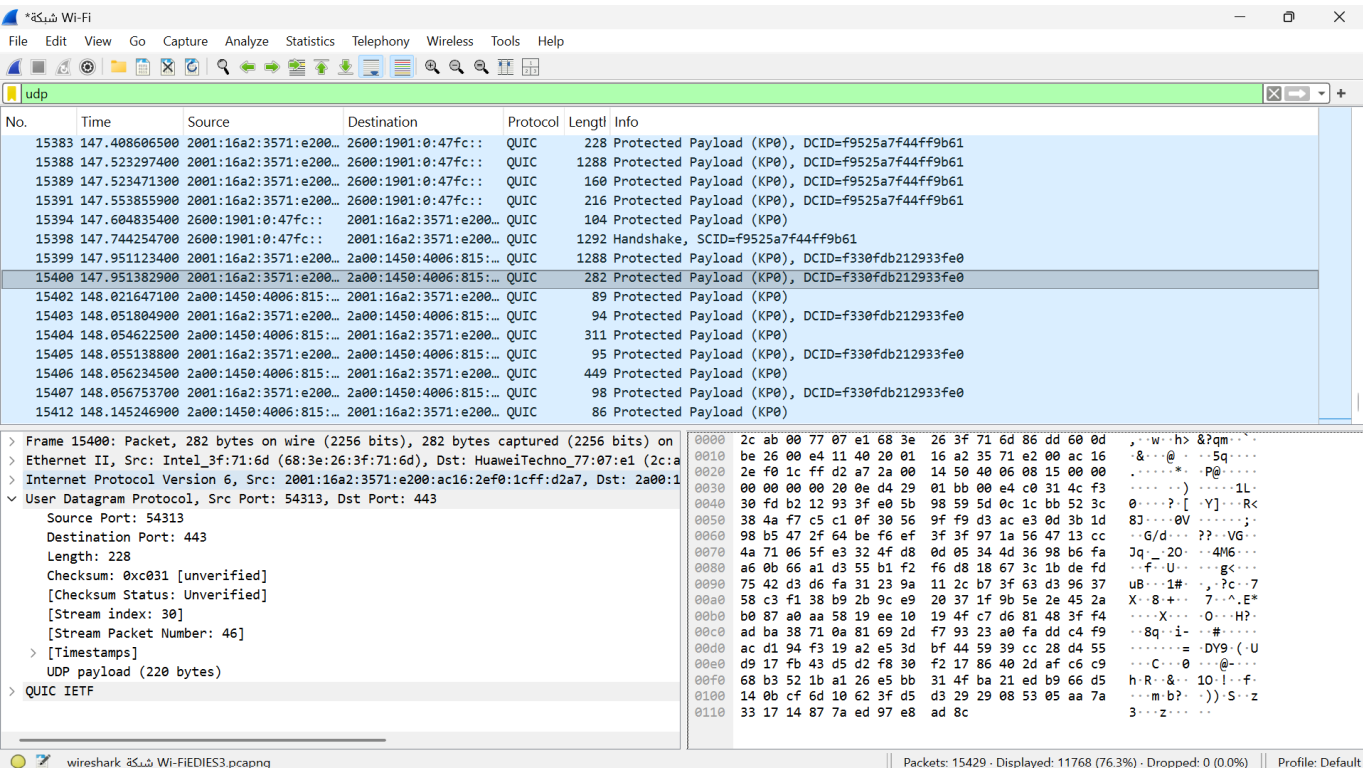
**Wireshark - Follow TCP Stream (tcp.flags.fin == 1) - شبكة Wi-Fi**

759 23.564050500 2a00:1450:4025:2405::88 2001:16a2:3571:e200::ac TCP 74 443 → 51000 [FIN, ACK] Seq=74 Ack=3 Win=1040 Len=0

Frame 759: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface  
Ethernet II, Src: HuaweiTechno\_77:07:e1 (2c:ab:00:77:07:e1), Dst: Intel\_3f:71:6d (68:3f:71:6d:00:00)  
Internet Protocol Version 6, Src: 2a00:1450:4025:2405::88, Dst: 2001:16a2:3571:e200::ac  
Transmission Control Protocol, Src Port: 443, Dst Port: 51000, Seq: 74, Ack: 3, Len: 0

## Part 3: Capturing and Analyzing UDP Traffic

### Screenshot 5 – UDP Packets



## Part 4: TCP and UDP Comparison

### Task 1: Reliability, Connection Establishment, Data Integrity, and Ordering

| Item                                     | TCP or UDP | Reason                                                                                                                            |
|------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Reliability and Connection Establishment | TCP        | TCP establishes a connection using the three-way handshake and provides acknowledgments and retransmission for reliable delivery. |
| Data Integrity and Ordering              | TCP        | TCP uses sequence numbers, acknowledgments, error checking, and retransmission to deliver data correctly and in order.            |

### Task 2: Use Cases and Performance

|             | TCP                                                                                                       | UDP                                                                                                |
|-------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Use cases   | Web browsing (HTTP/HTTPS), email, file transfer, remote login, and online banking.                        | Video streaming, VoIP, online gaming, DNS, and other real-time applications.                       |
| Performance | Usually slower because of connection setup, acknowledgments, retransmission, ordering, and error control. | Usually faster because it is connectionless, has a smaller header, and has less protocol overhead. |